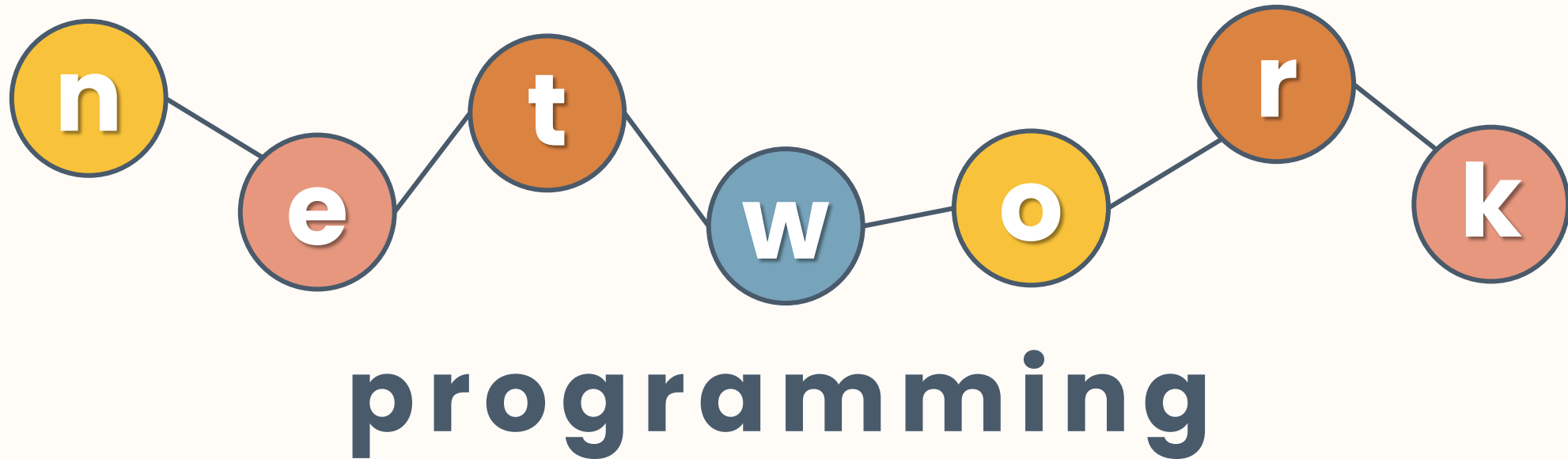


Web Services, Middleware

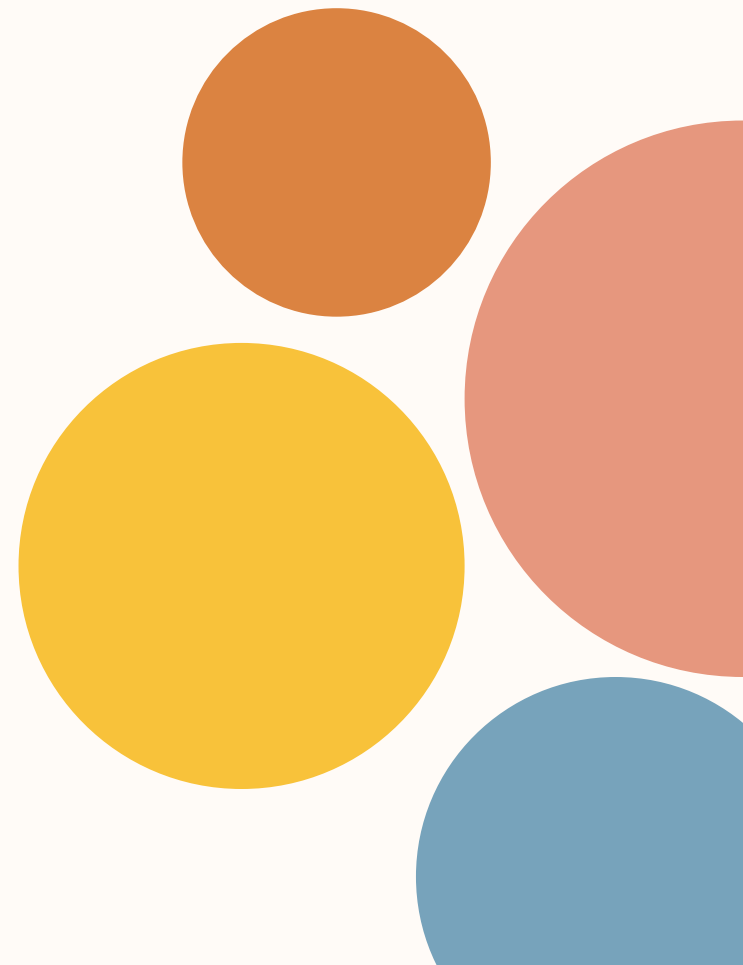


IT 332: INTEGRATIVE PROGRAMMING AND TECHNOLOGIES

Prepared by: **Ms. FATIMA MARIE P. AGDON**

In this presentation

You will be able to describe the web services and middleware, identify the parts of network programming, and, modify a low-level data communications message and queuing services.



Web Services and Middleware



Middleware serves as medium for program-to-program interactions.

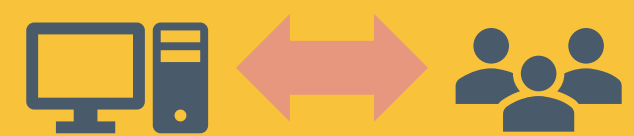


The **web service** is for program-to-user interactions.

Web Services and Middleware



Middleware serves as medium for program-to-program interactions.



The **web service** is for program-to-user interactions.



This allows companies to reduce the cost of doing e-business.

Web services model built on emerging standards such as:

HTML

XML

SOAP

WSDL

UDDI

are developed to:

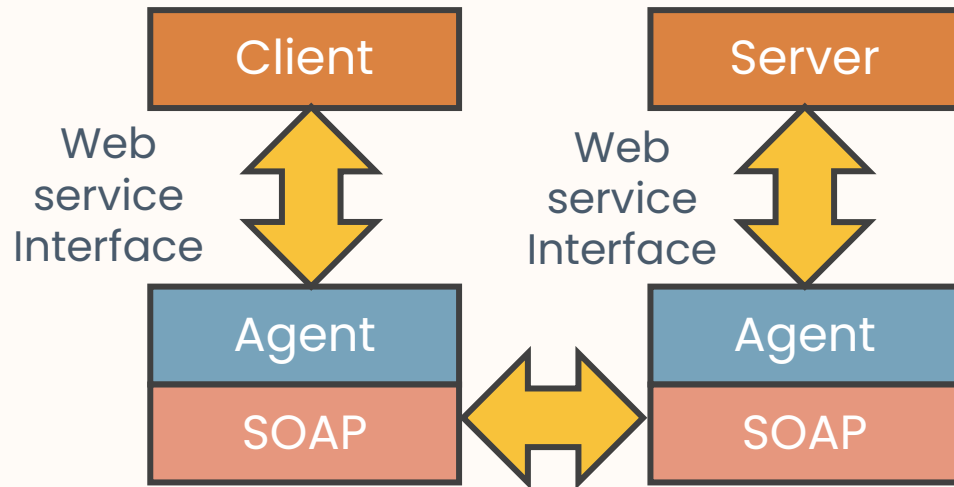


Distribute an object and serve it to various users in the web environments used in the server situations



Solving the web-scalability problem of the other distributed object technologies at the same time.

Web Services and Middleware



1. Client that wants to be serviced should first find the supported services from the **pre-existing registry** before compiling the code.
2. The client gains the **Web Service Description Language** (WSDL) that a server previously registered.
 - From the WSDL, the client knows the service provider location and the parameters for the found method.
3. After binding the described service during the compile time, it **calls the local agent whenever the client invokes a method call**, and the local agent delivers it to the server-side agent through Simple Object Access Protocol (SOAP) over HTTP, FTP, SMTP, and TCP during the runtime.
4. The **server-side agent activates the appropriate object**, and delivers the calls to the object.

Network Programming



Network Programming



NETWORK

NODE

PACKET

PROTOCOL

IP

TCP

UDP

PORTS

INTERNET

SOCKET

IP ADDRESS

Network Programming



NETWORK

NODE

PACKET

PROTOCOL

IP

TCP

UDP

PORTS

INTERNET

SOCKET

IP ADDRESS

NETWORK

It is a collection of computers and other devices that can send data to and receive data from each other.

Network Programming



NETWORK

NODE

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IP

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PORTS

INTERNET

SOCKET

IP ADDRESS

NODE

It refers to each machine on a network. Computers are mostly considered as nodes, but also printers, routers, bridges, gateways are considered as node.

Network Programming



NETWORK

NODE

PACKET

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IP

TCP

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INTERNET

SOCKET

IP ADDRESS

PACKET

It refers to the data traveling on the network is broken into chunks. Each packet is handled separately and contains information about who sent it and where it's going.

Network Programming



NETWORK

NODE

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PROTOCOL

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SOCKET

IP ADDRESS

PROTOCOL

It is a precise set of rules defining how computers communicate: the format of addresses, how data is split into packets, etc.

Network Programming



NETWORK

NODE

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INTERNET

SOCKET

IP ADDRESS

INTERNET PROTOCOL

It was designed to allow multiple routes between any two points and to route packets of data around damaged routers.

Network Programming



NETWORK

NODE

PACKET

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IP ADDRESS

TRANSMISSION CONTROL PROTOCOL

It is a standard that defines how to establish and maintain a network conversation by which applications can exchange data.

Network Programming



NETWORK

NODE

PACKET

PROTOCOL

IP

TCP

UDP

PORTS

INTERNET

SOCKET

IP ADDRESS

USER DATAGRAM PROTOCOL

It is an unreliable protocol that does not guarantee that packets will arrive at their destination or that they will arrive in the same order they were sent.

Network Programming



NETWORK

NODE

PACKET

PROTOCOL

IP

TCP

UDP

PORTS

INTERNET

SOCKET

IP ADDRESS

PORTS

It is identified by a number between 1 and 65,535. Each port can be allocated to a particular service.

Network Programming



NETWORK

NODE

PACKET

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IP ADDRESS

INTERNET

It is the largest IP-based network for connecting machines together.

Network Programming



NETWORK

NODE

PACKET

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SOCKET

IP ADDRESS

SOCKET

Sockets are a means of using IP to communicate between machines, so sockets allow Java to interoperate with legacy systems by simply talking to existing servers using their predefined protocol.

Network Programming



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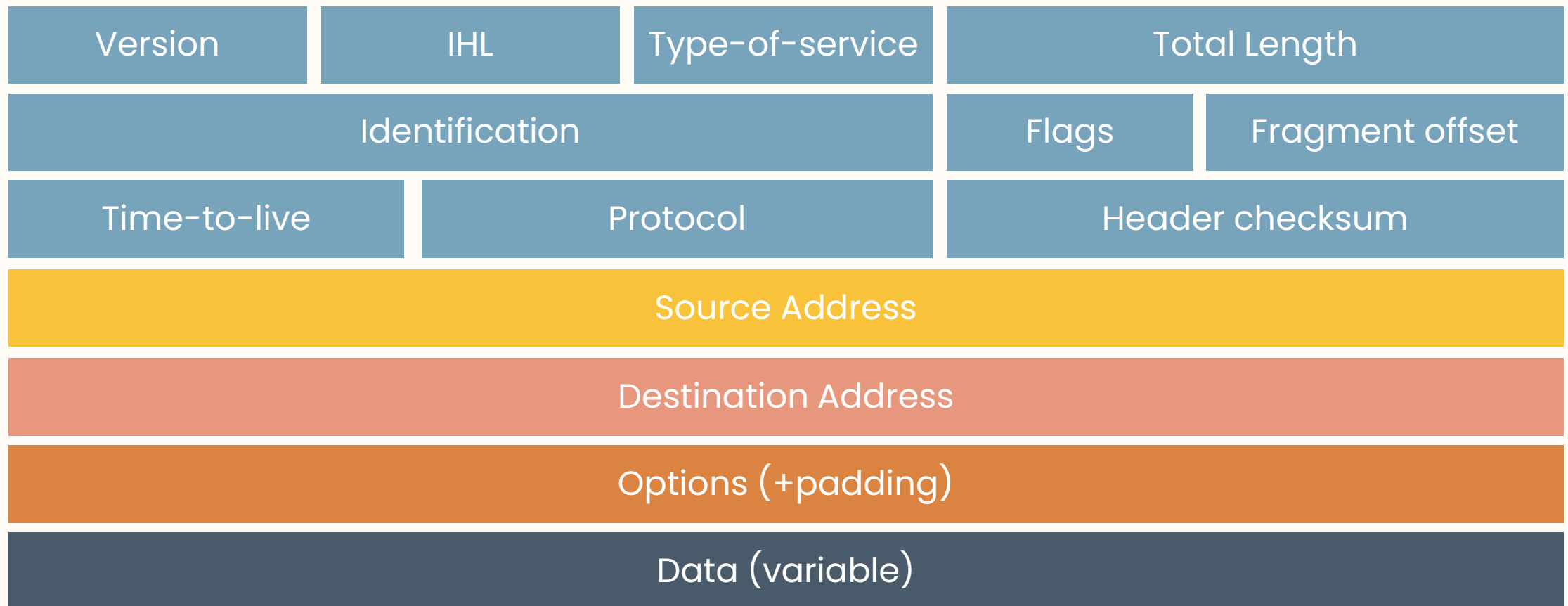
IP ADDRESS

Every network node has an address—a series of bytes that uniquely identify it. Internet addresses are manipulated in Java through the use of the `InetAddress` class. `InetAddress` takes care of the Domain Name System (DNS) look-up and reverselook-up;

Low-level Data Communication

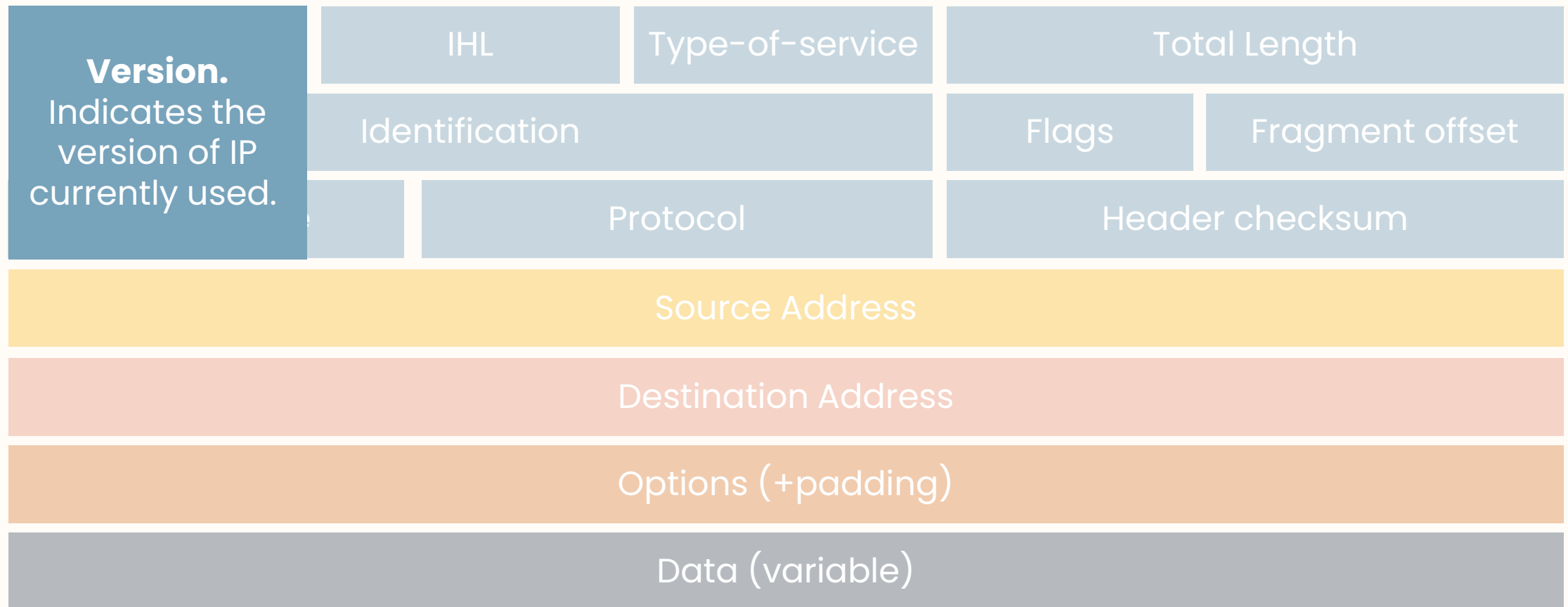
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The IP (Internet Protocol) is responsible for delivering packets of data between systems on the internet and specifies their format.



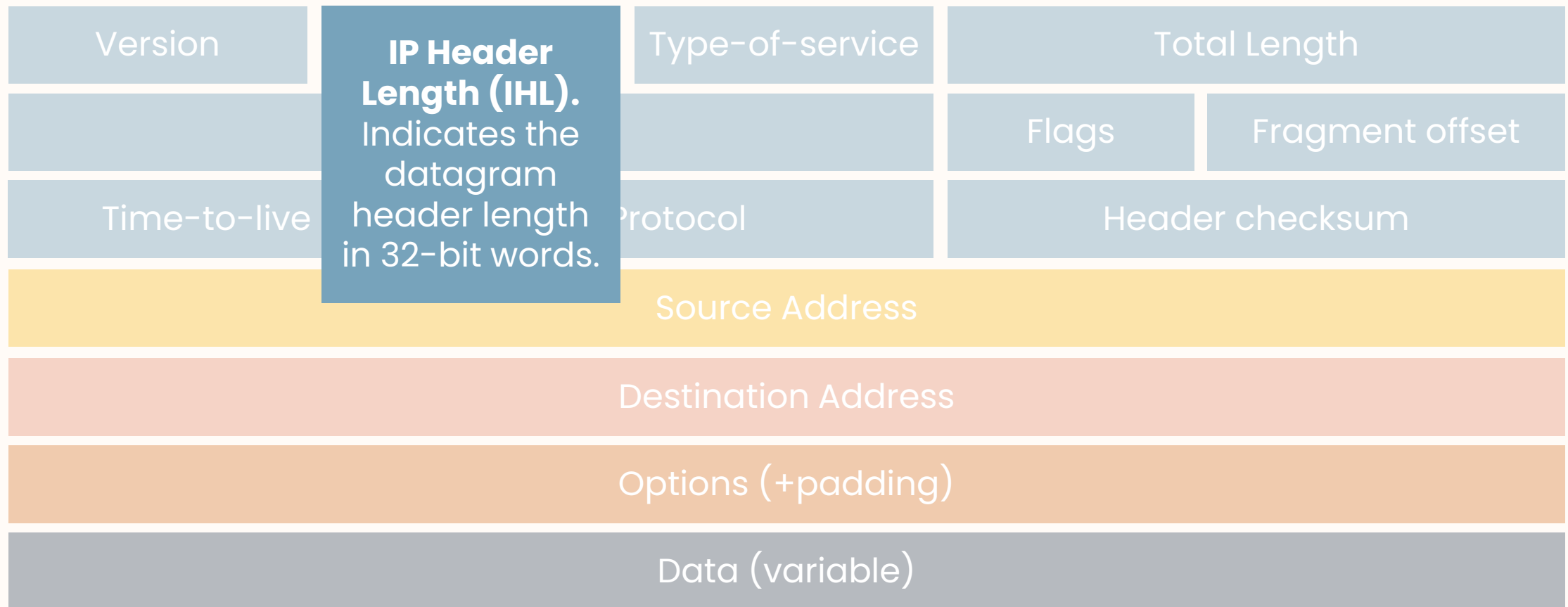
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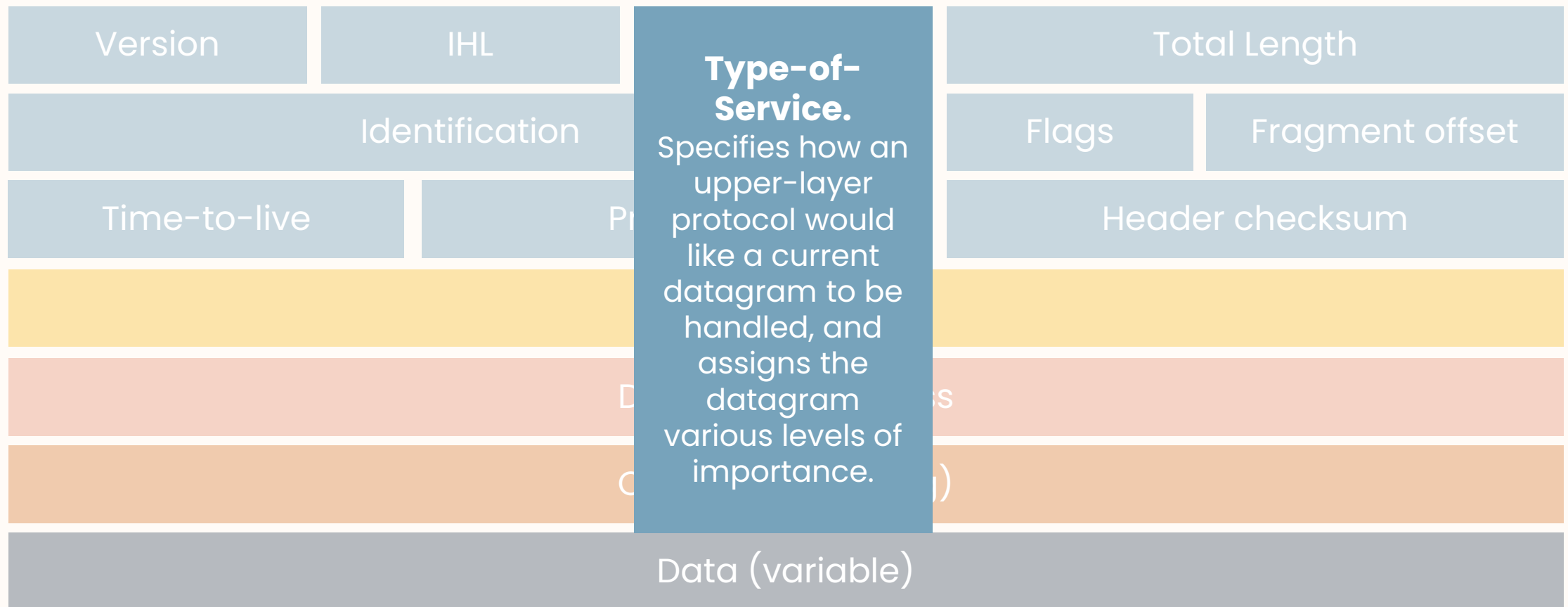
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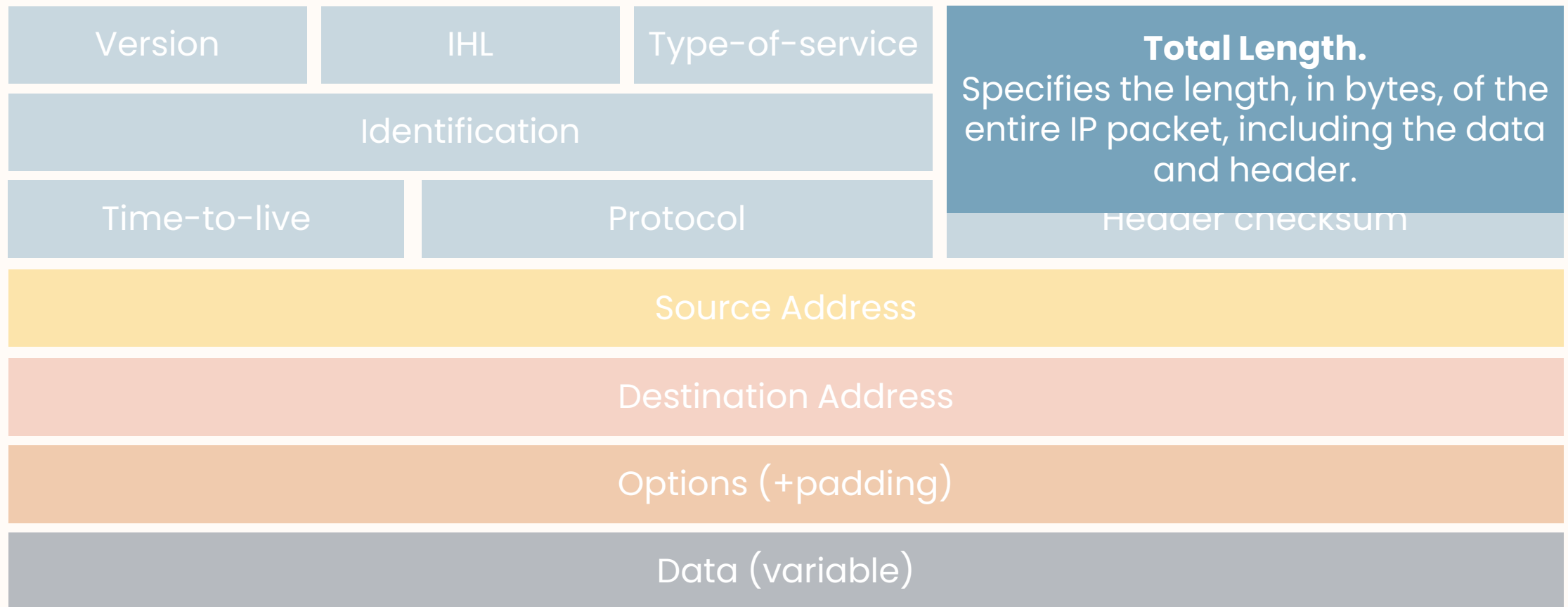
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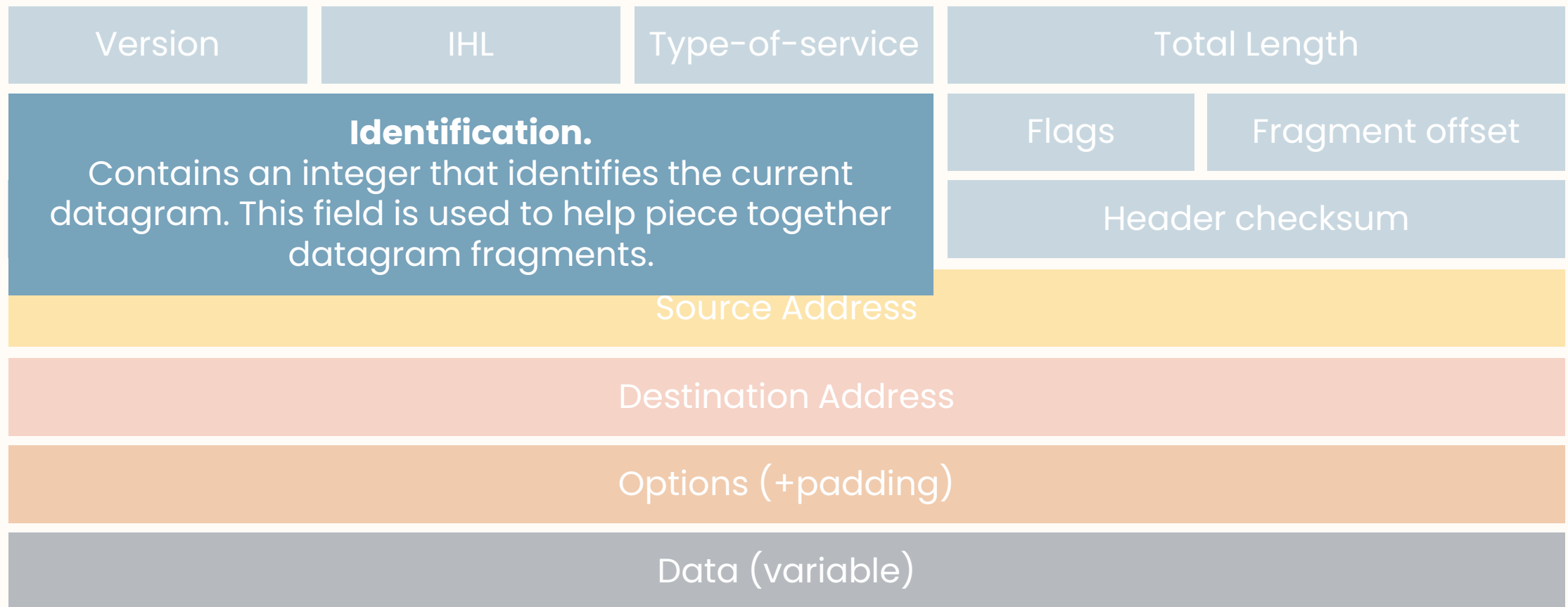
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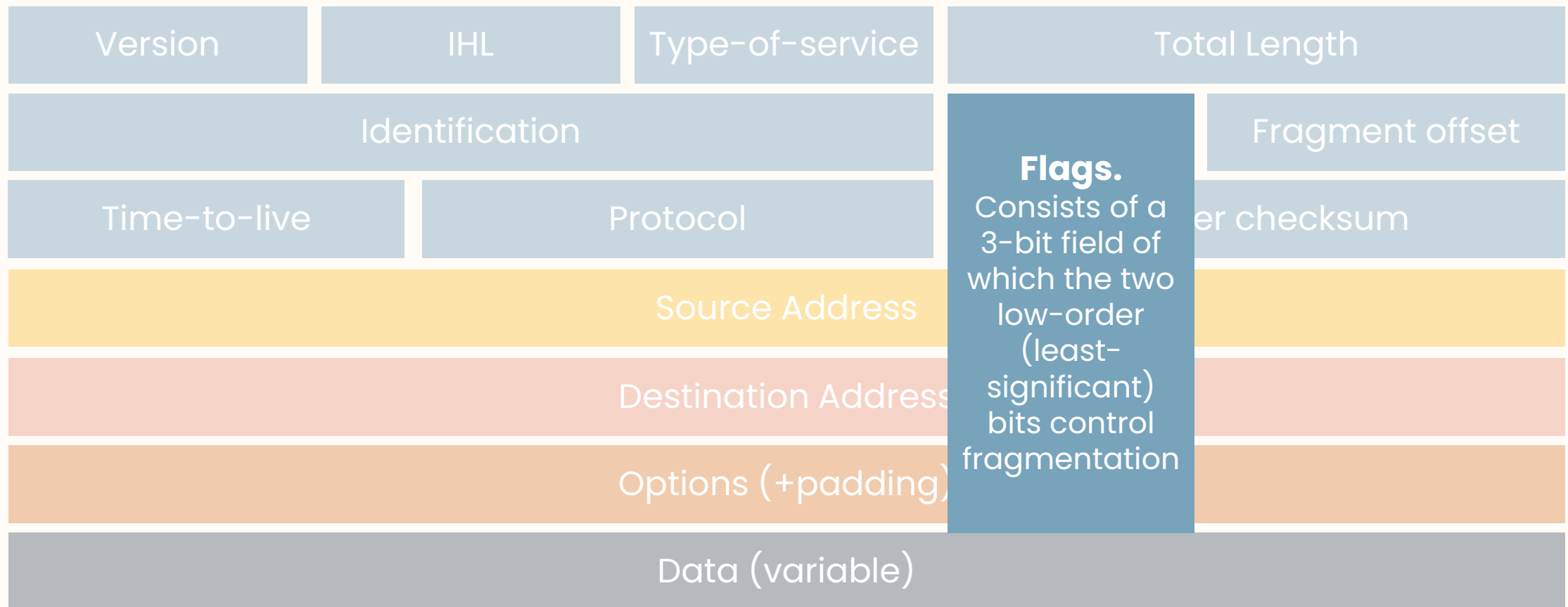
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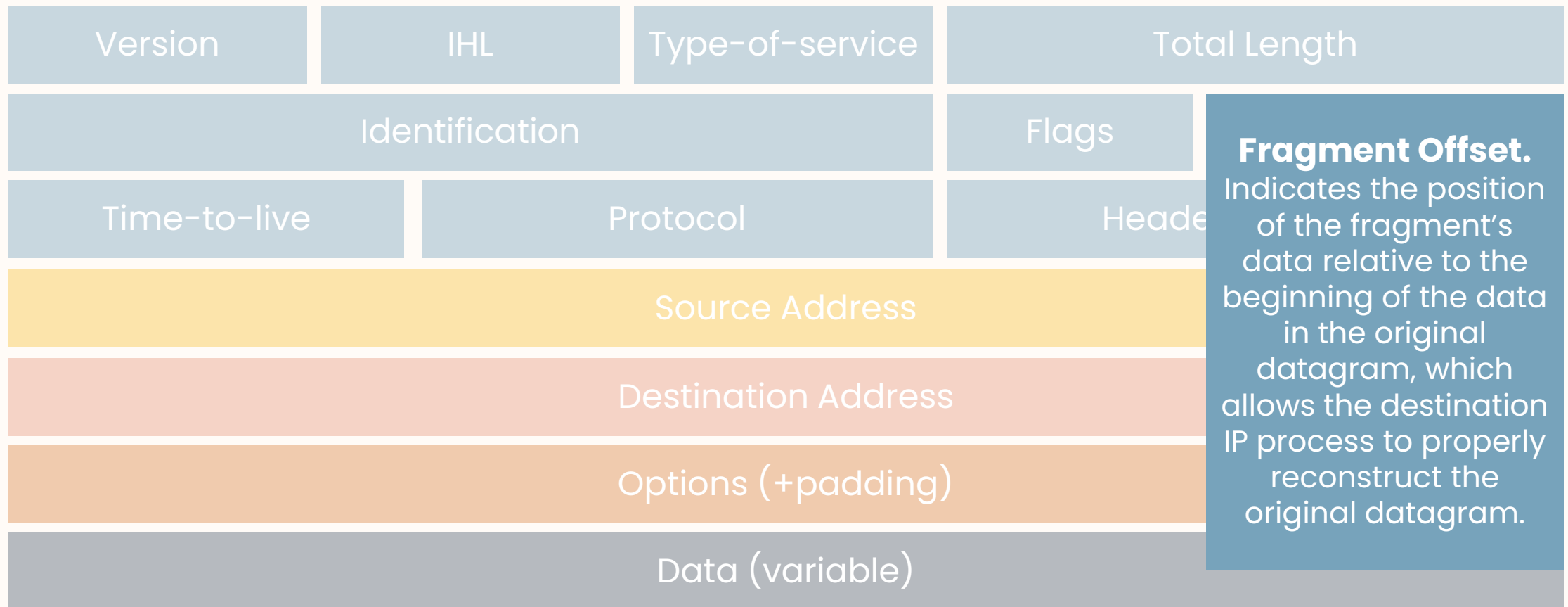
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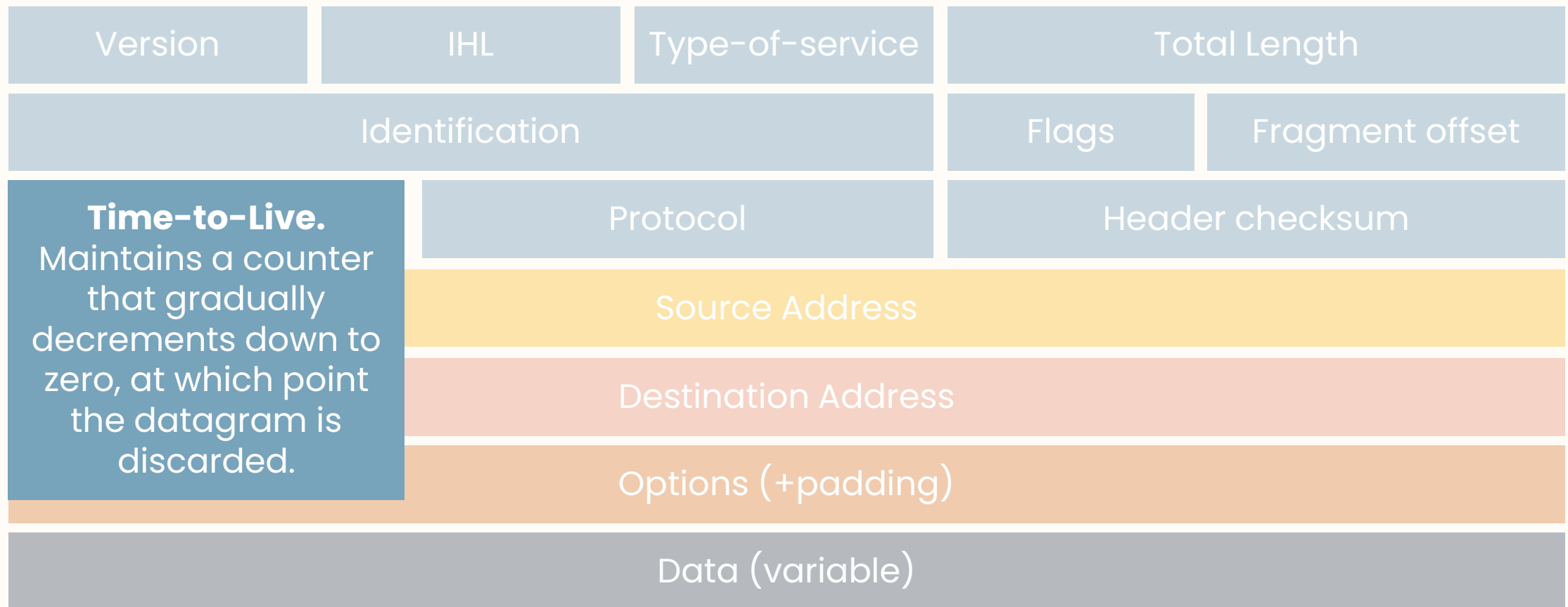
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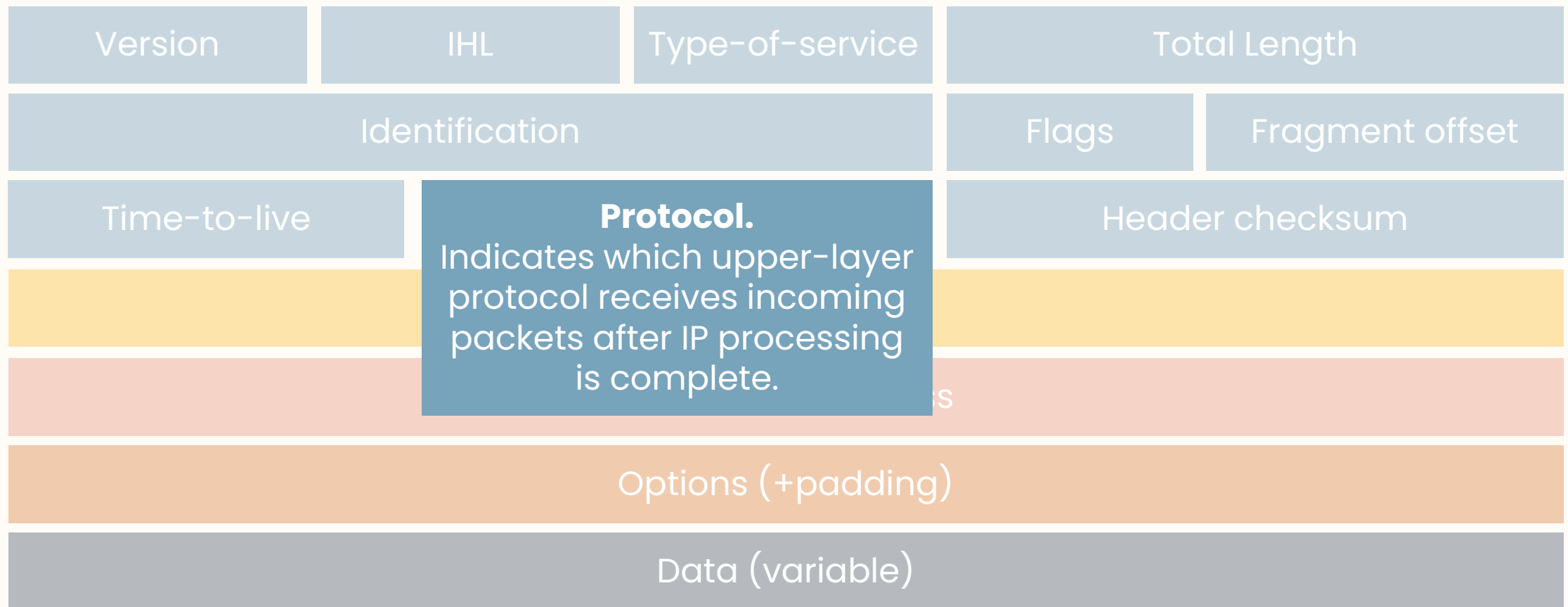
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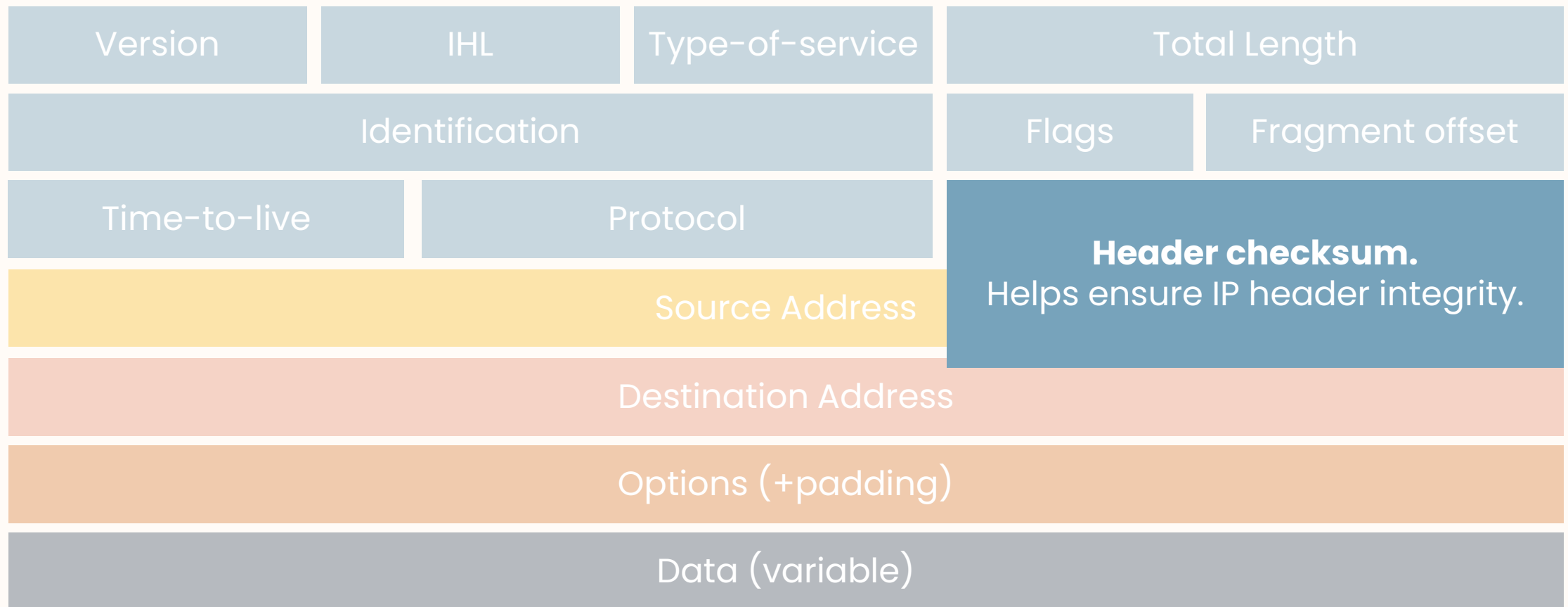
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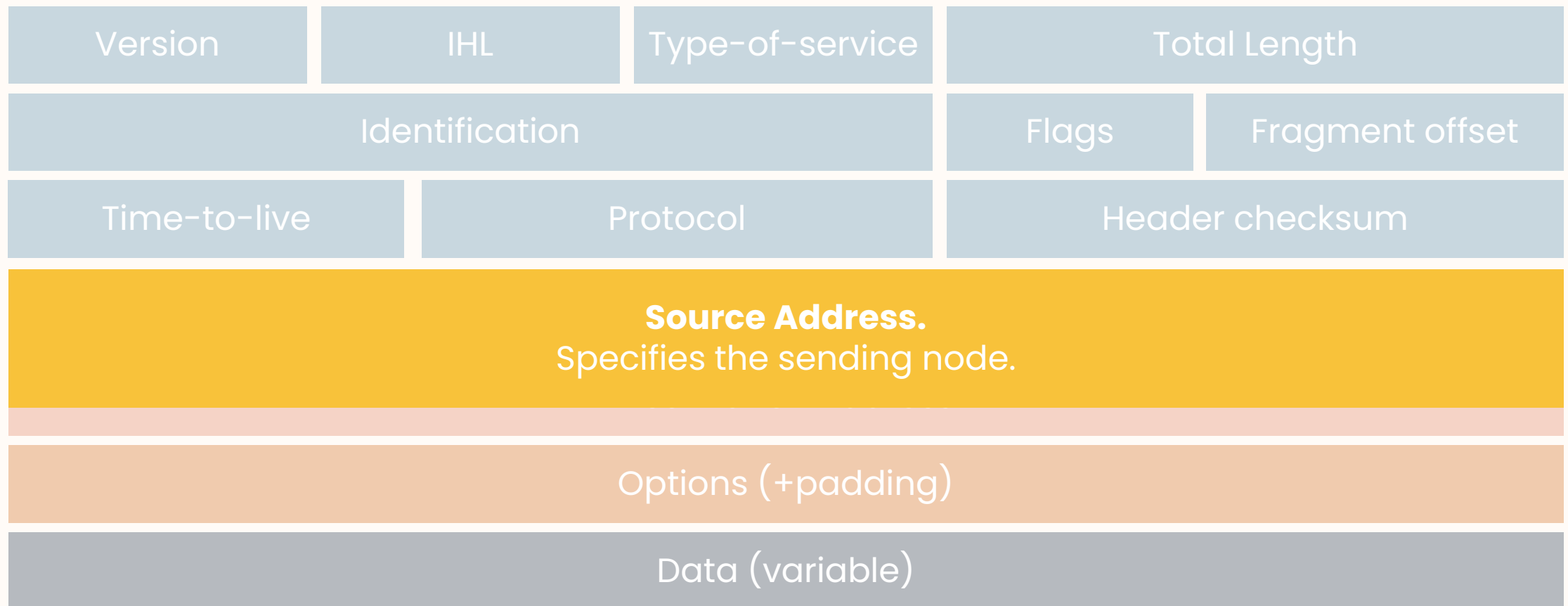
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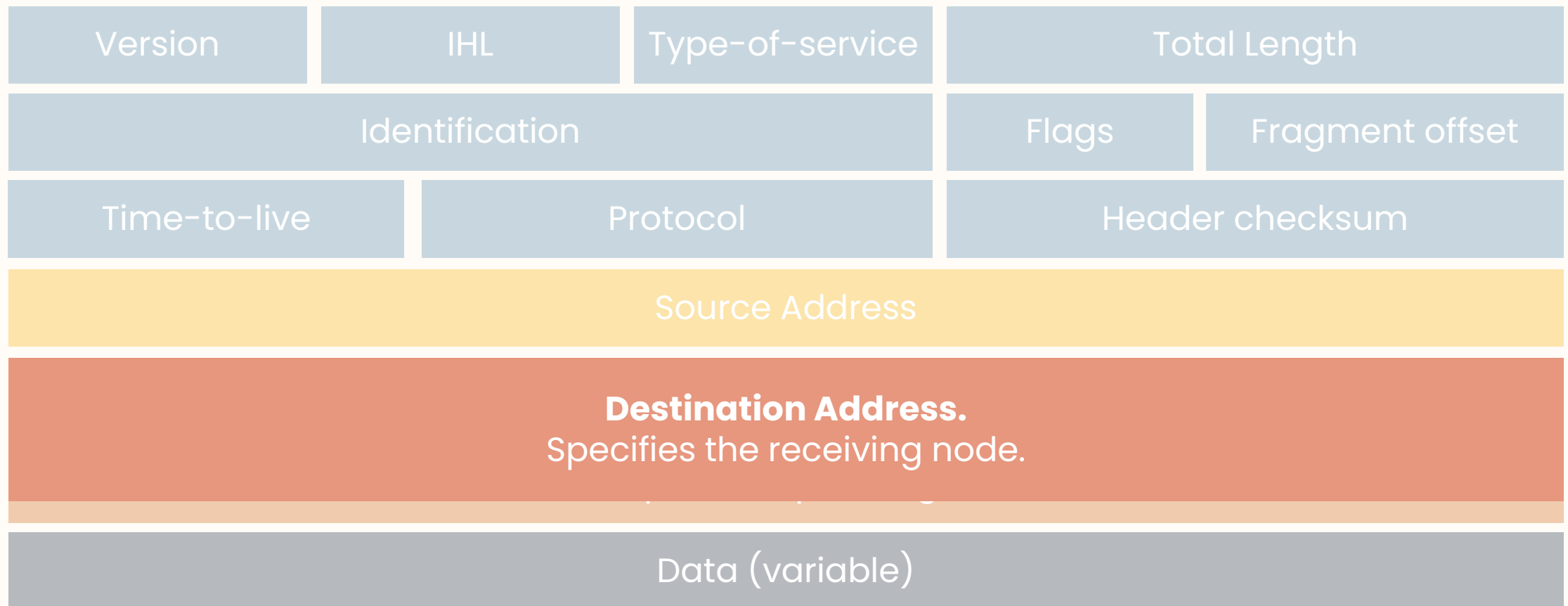
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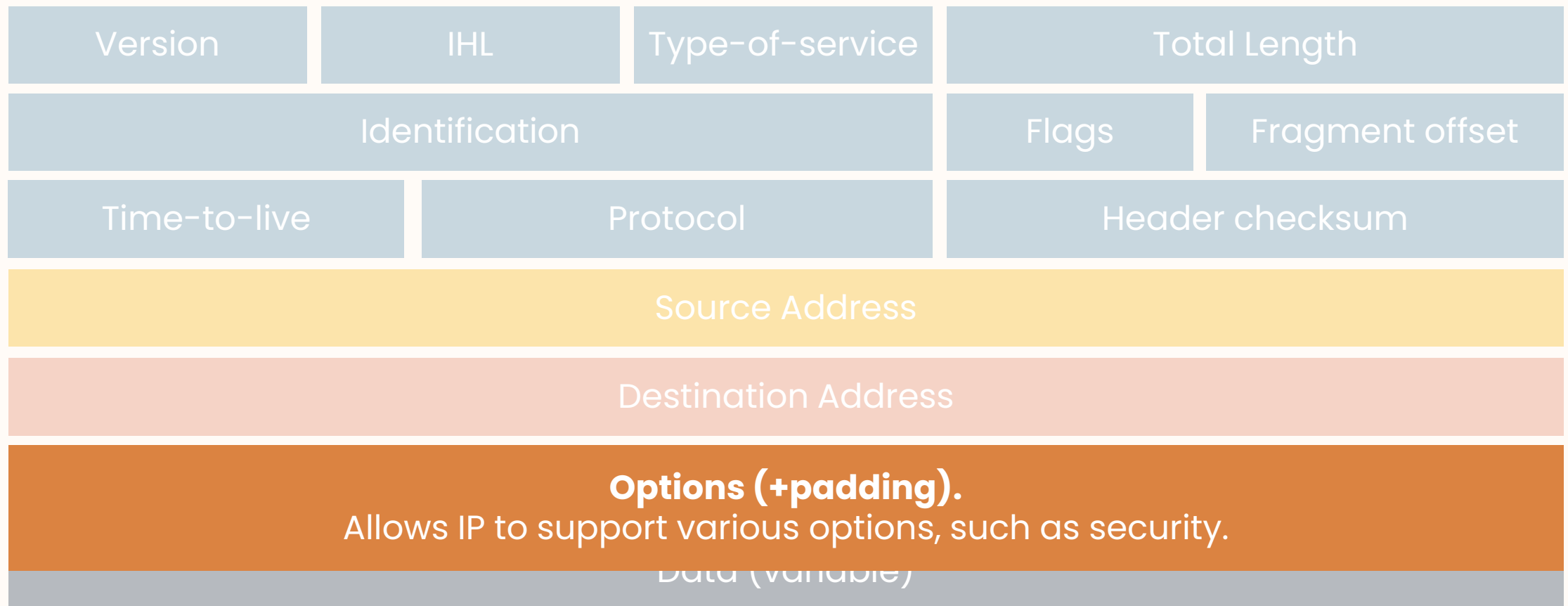
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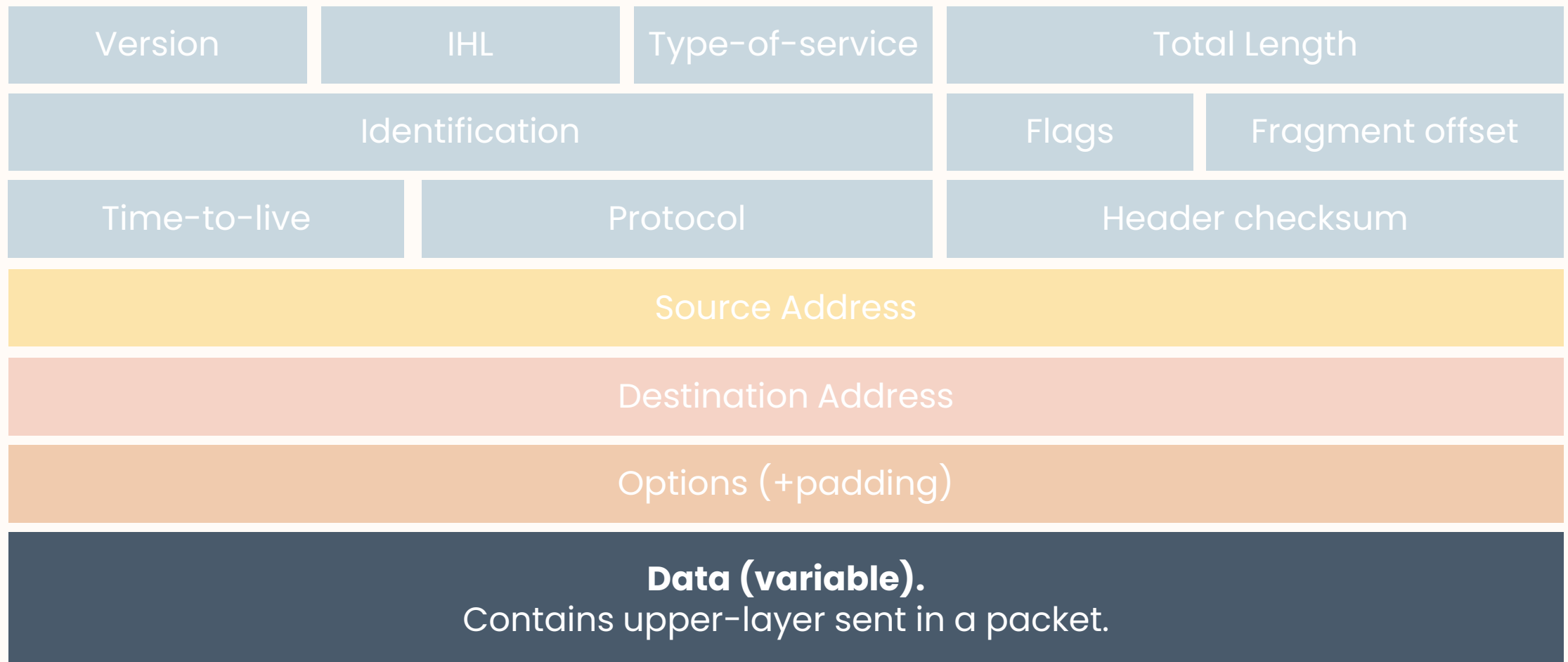
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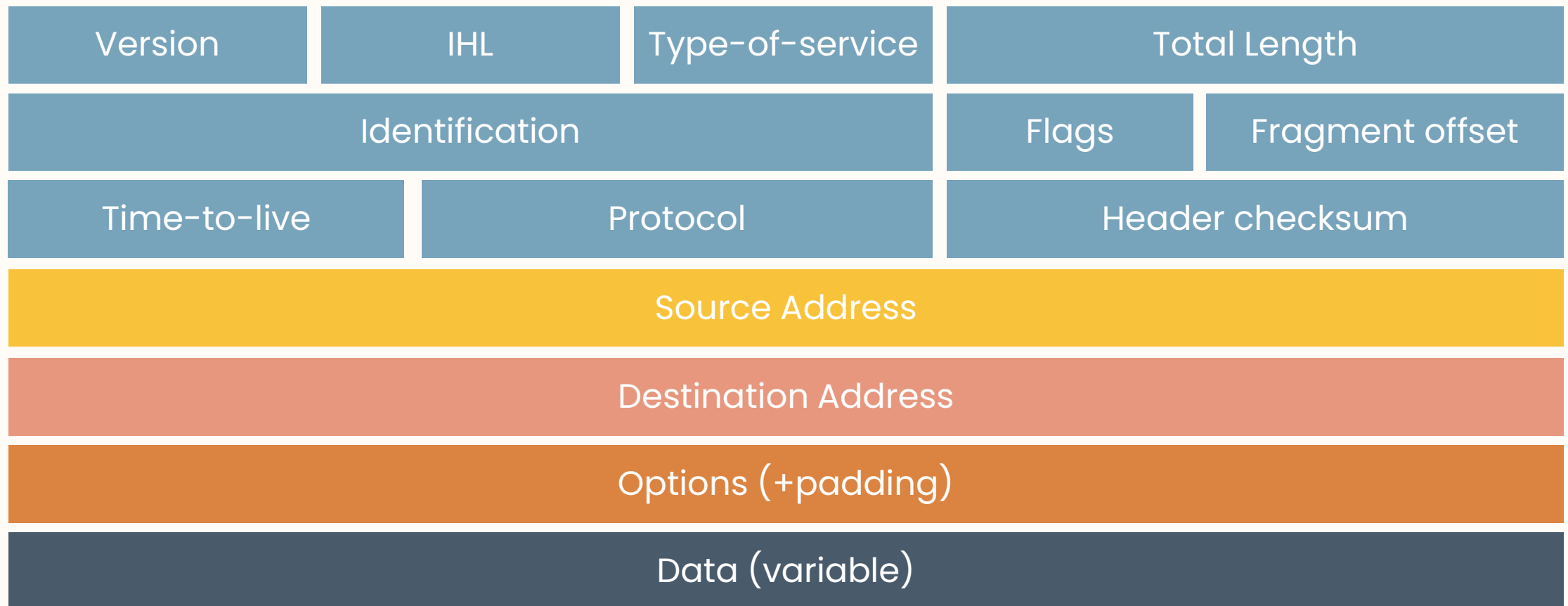
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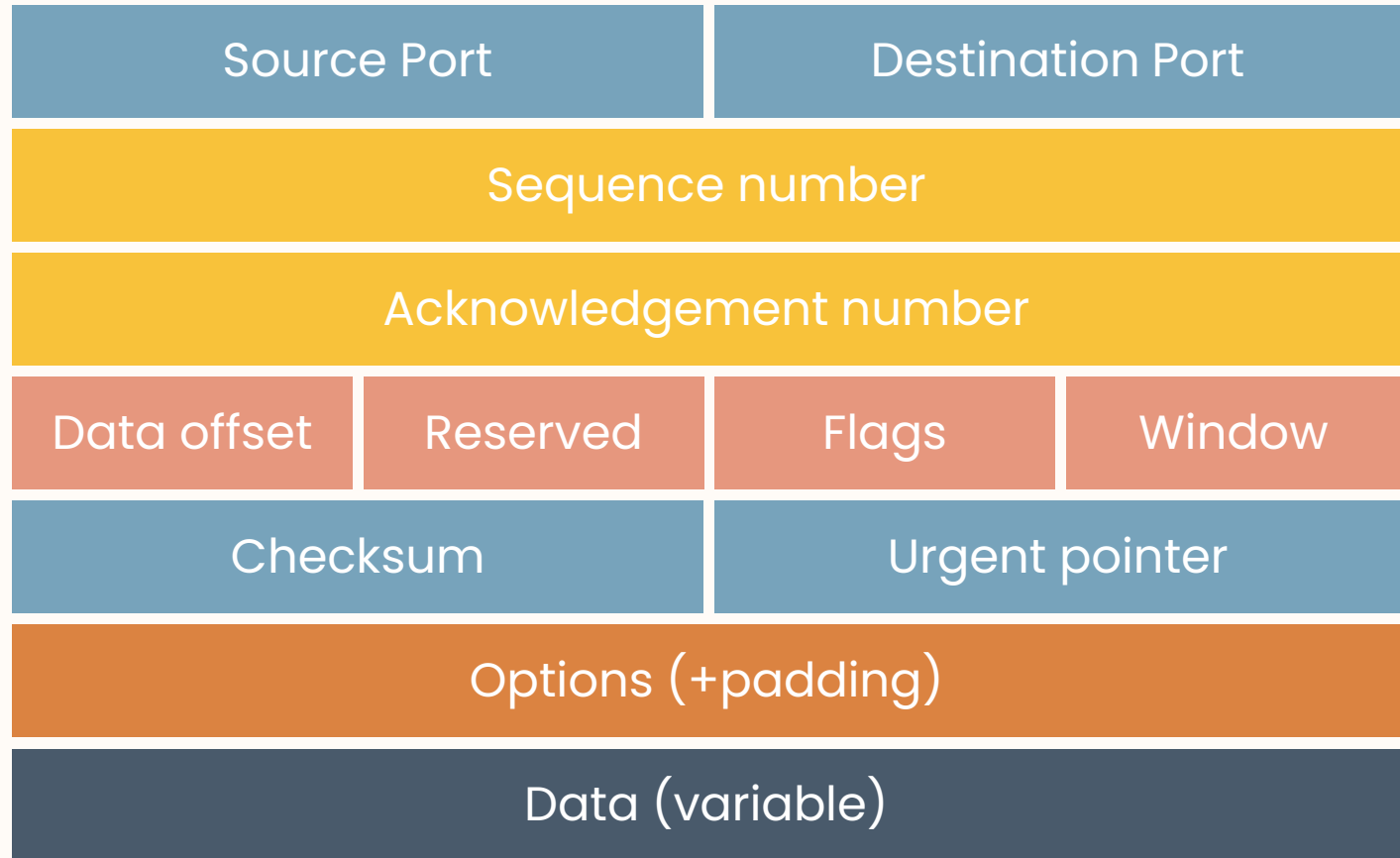
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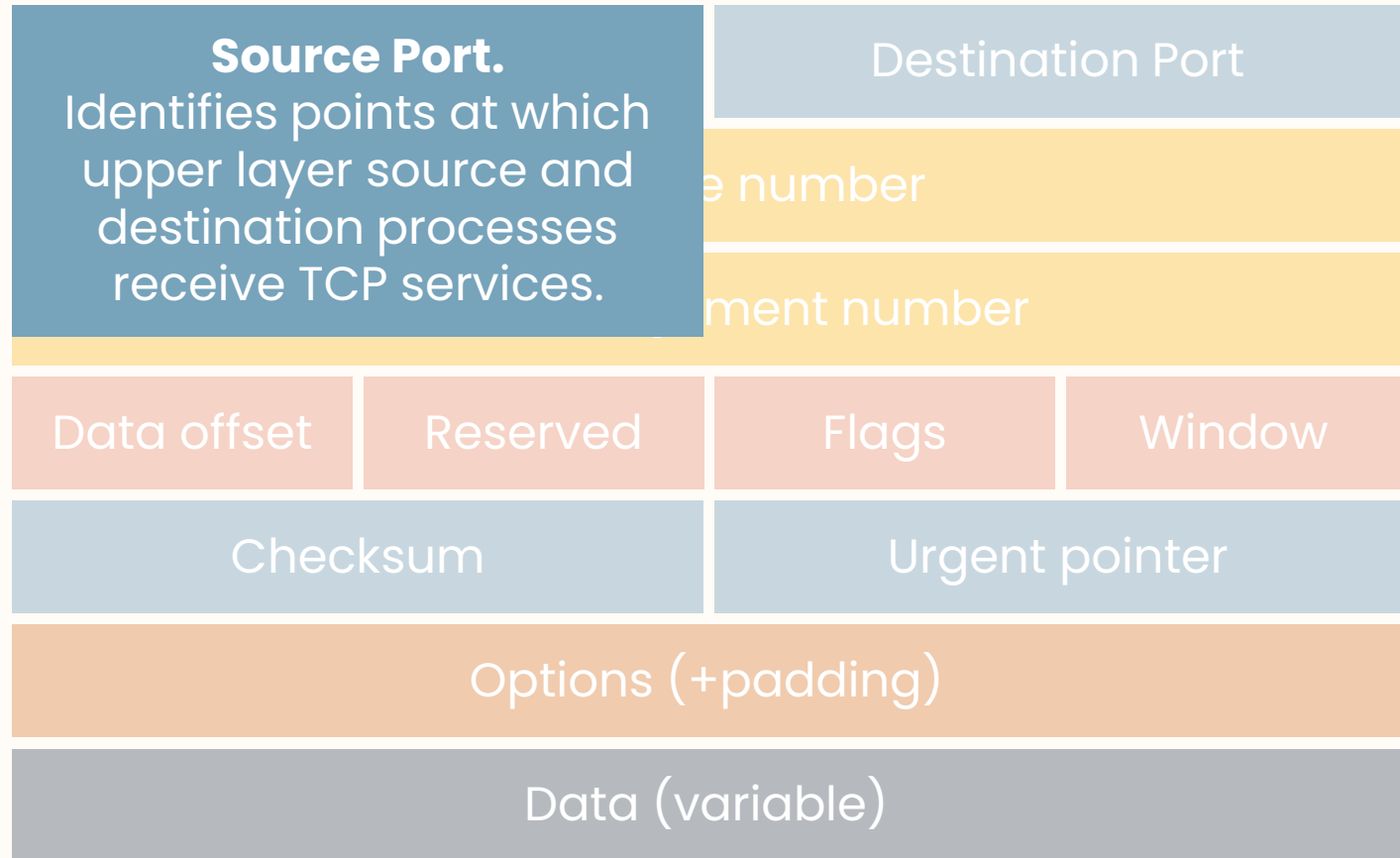
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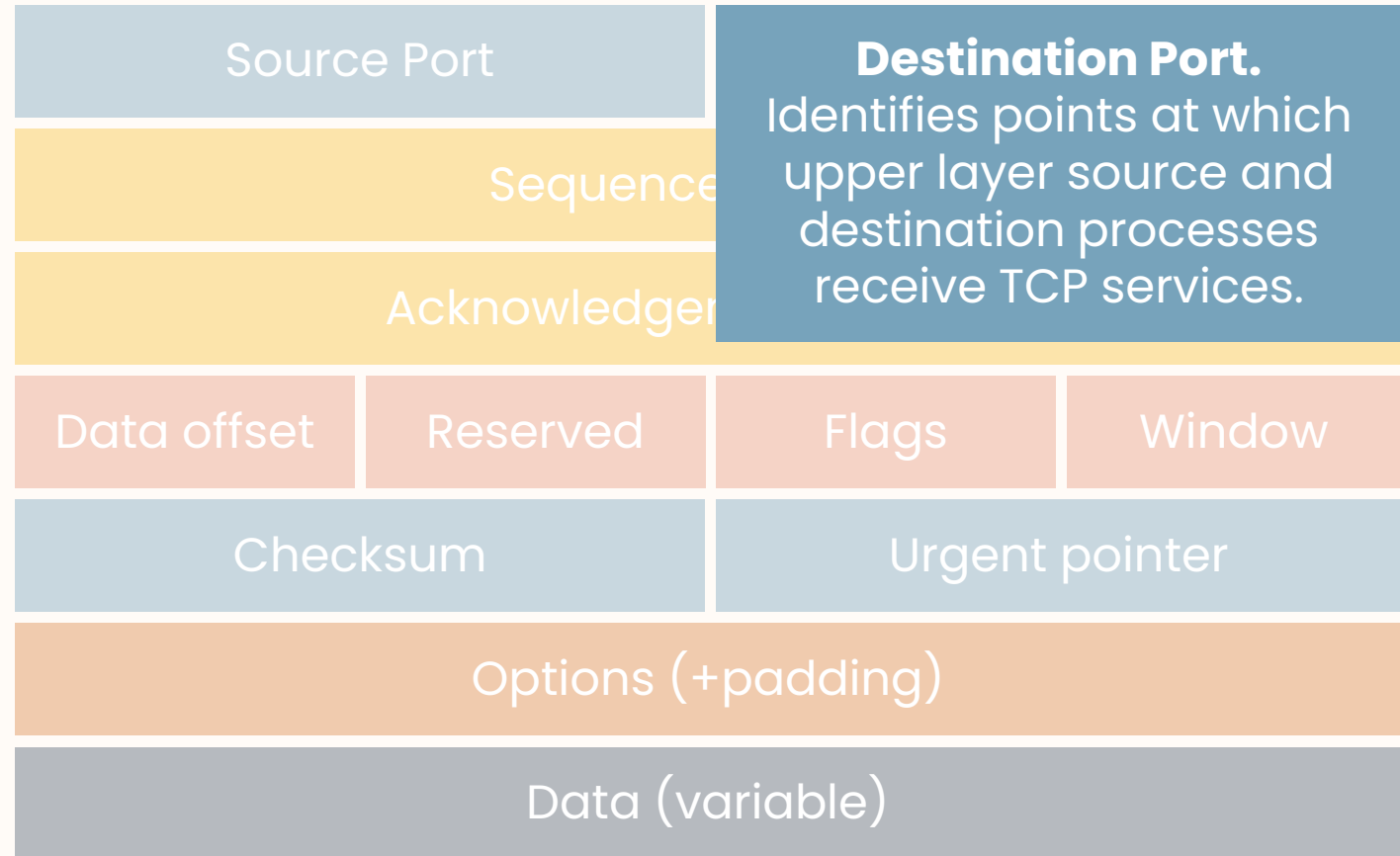
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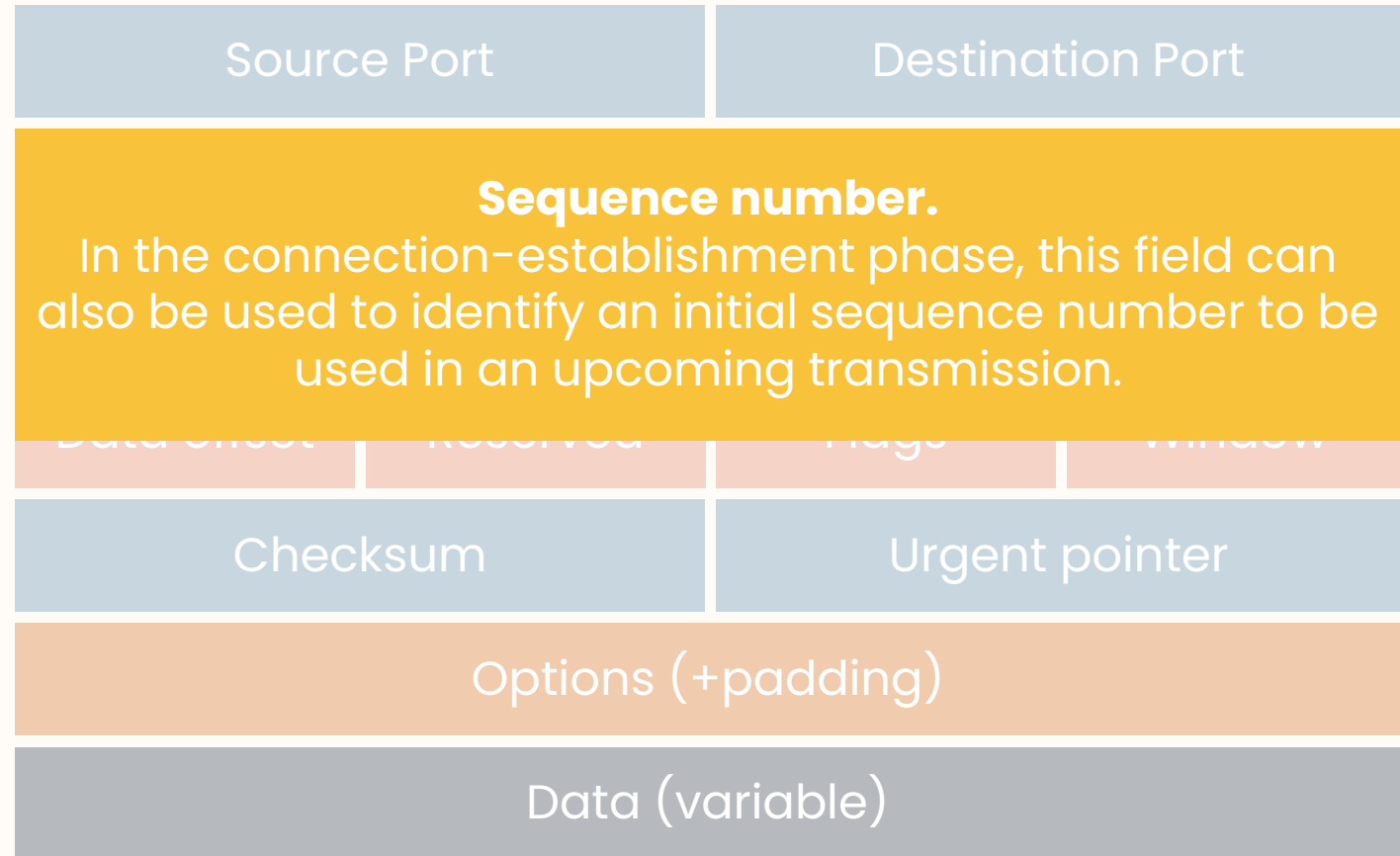
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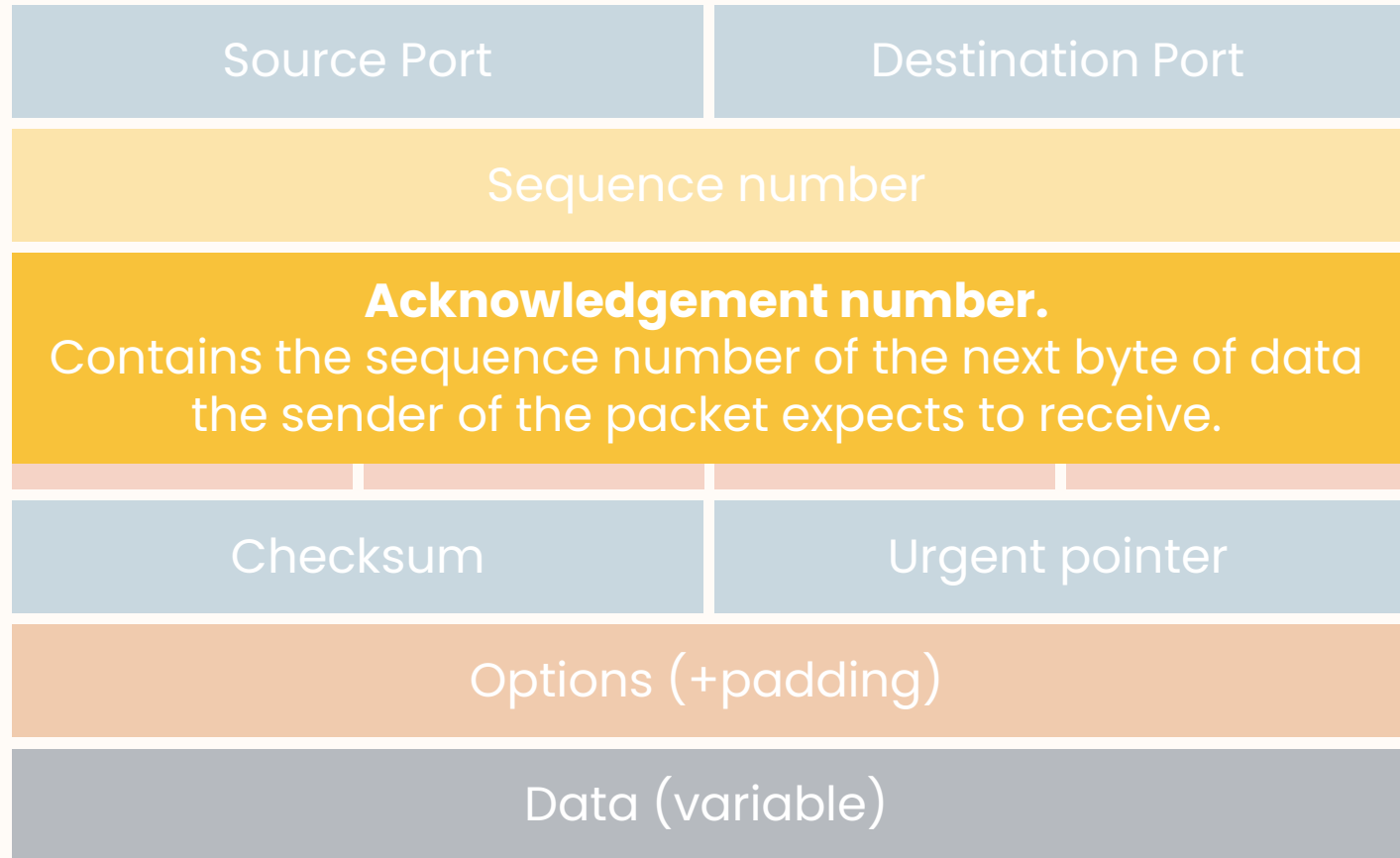
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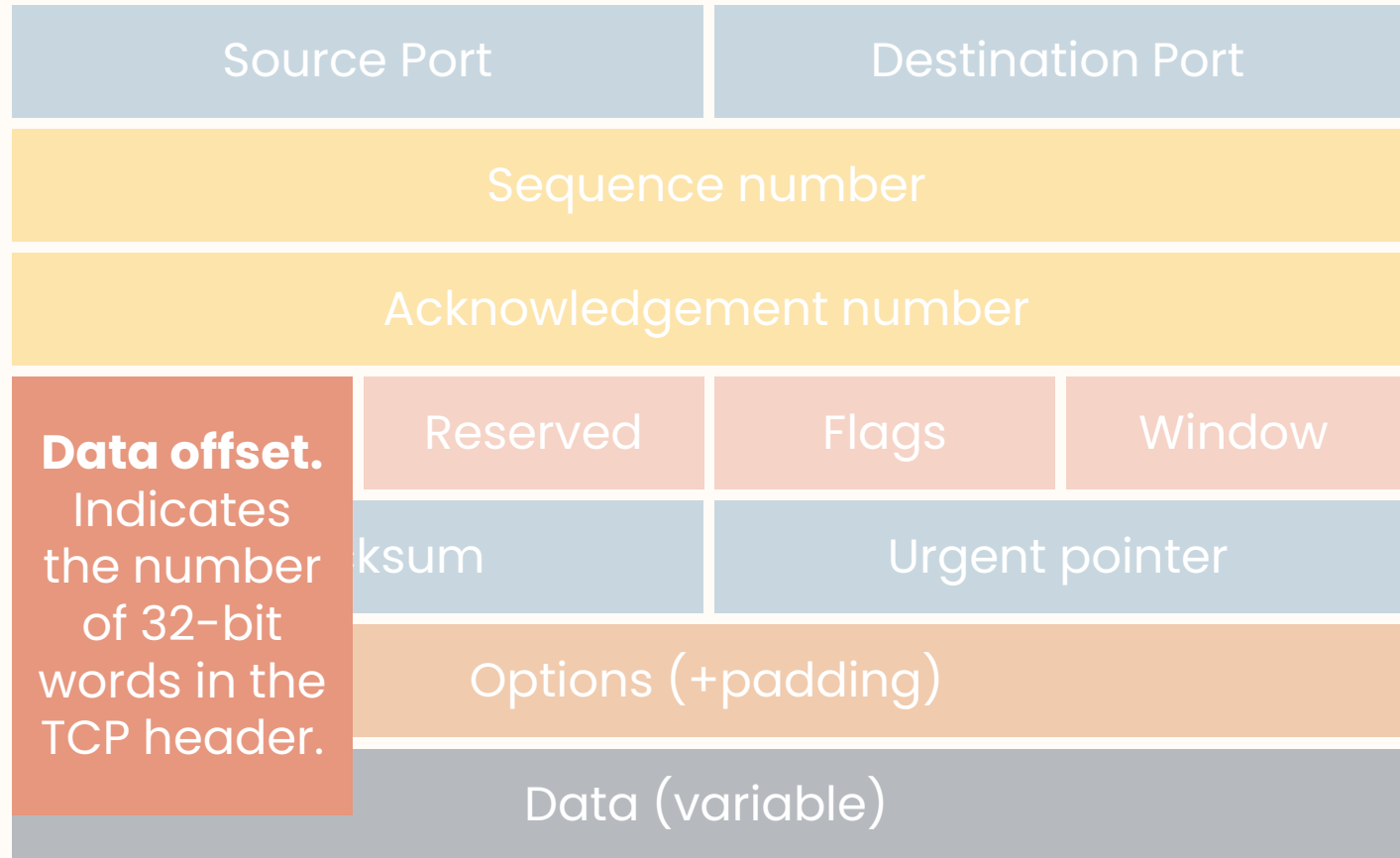
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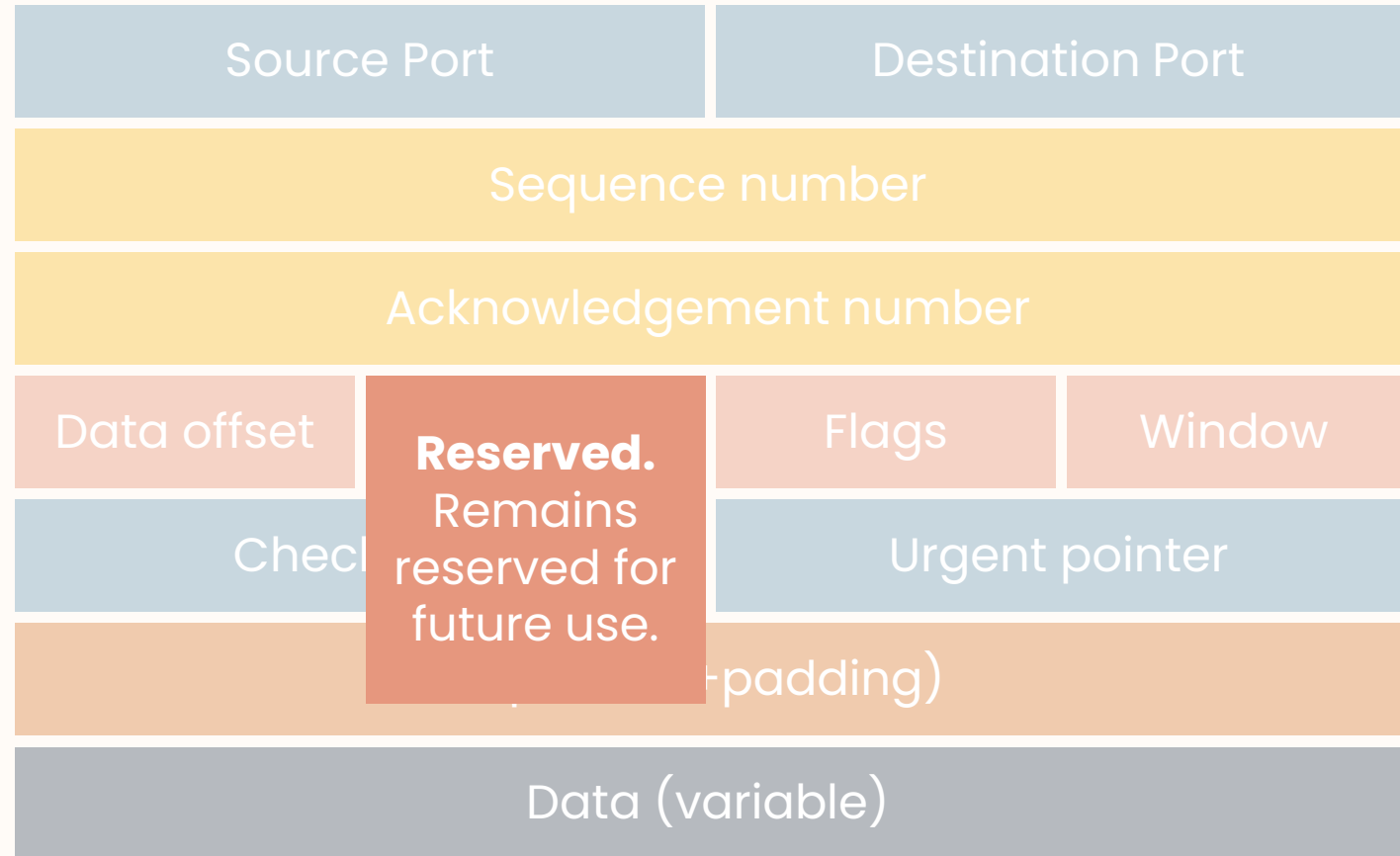
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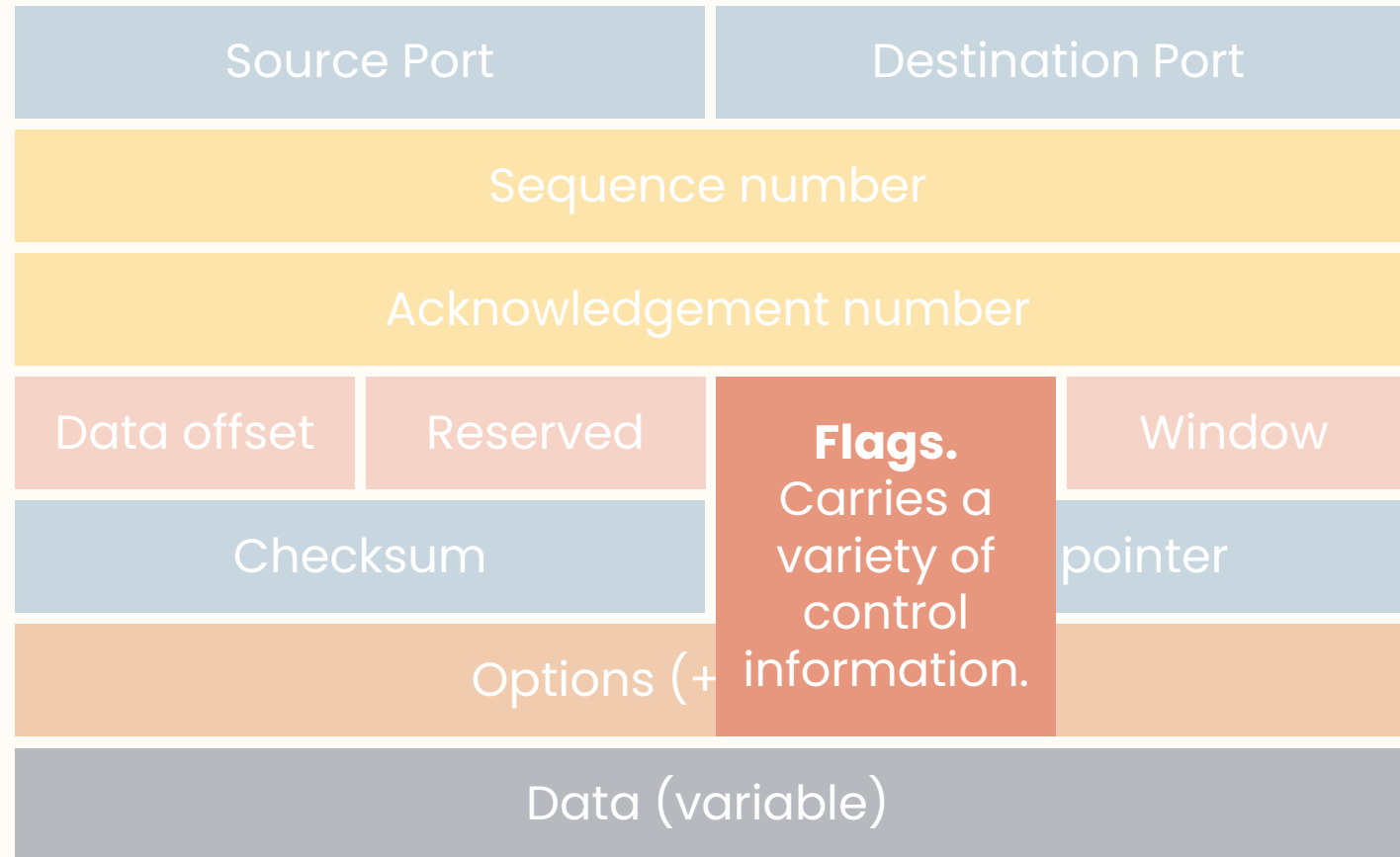
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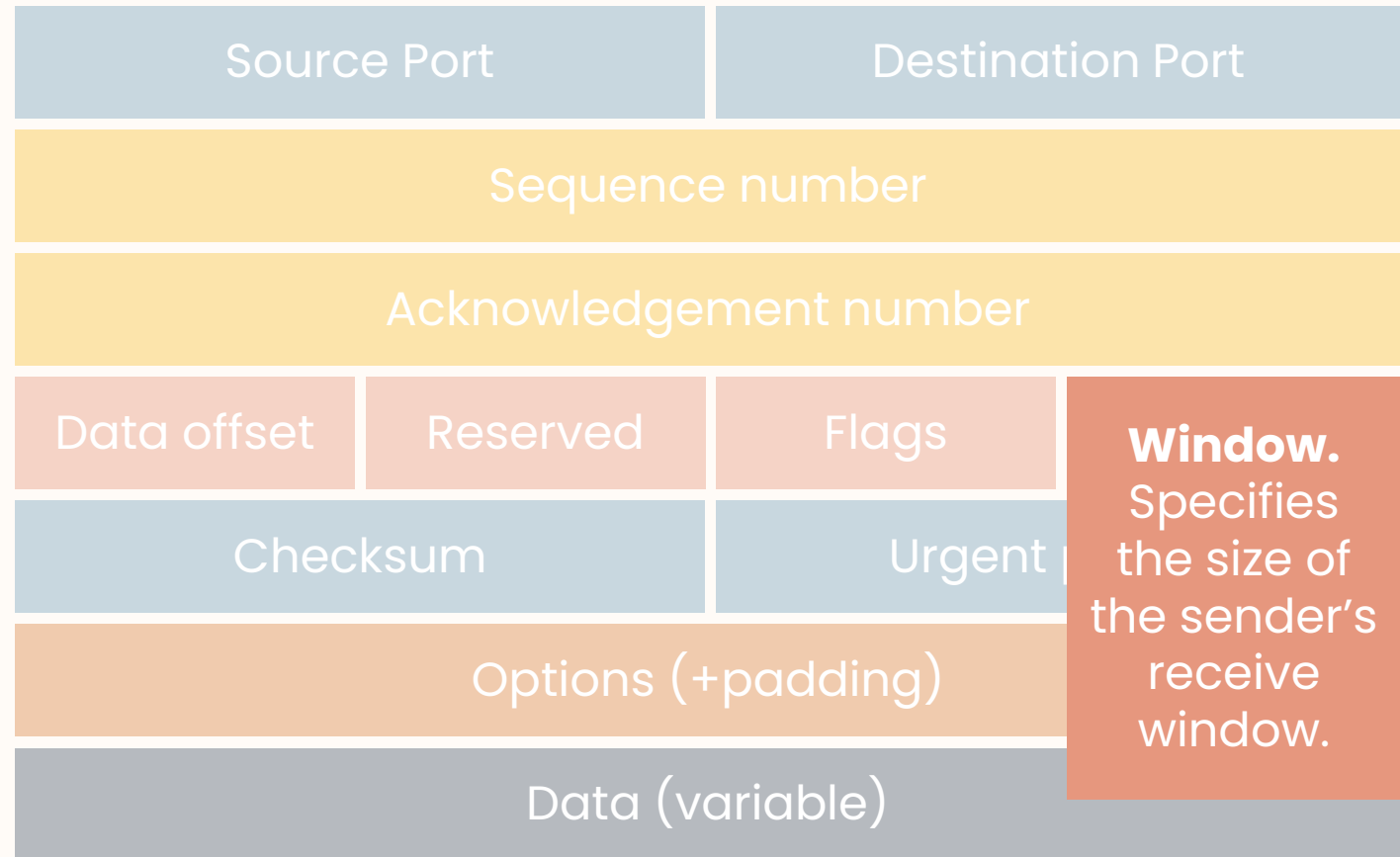
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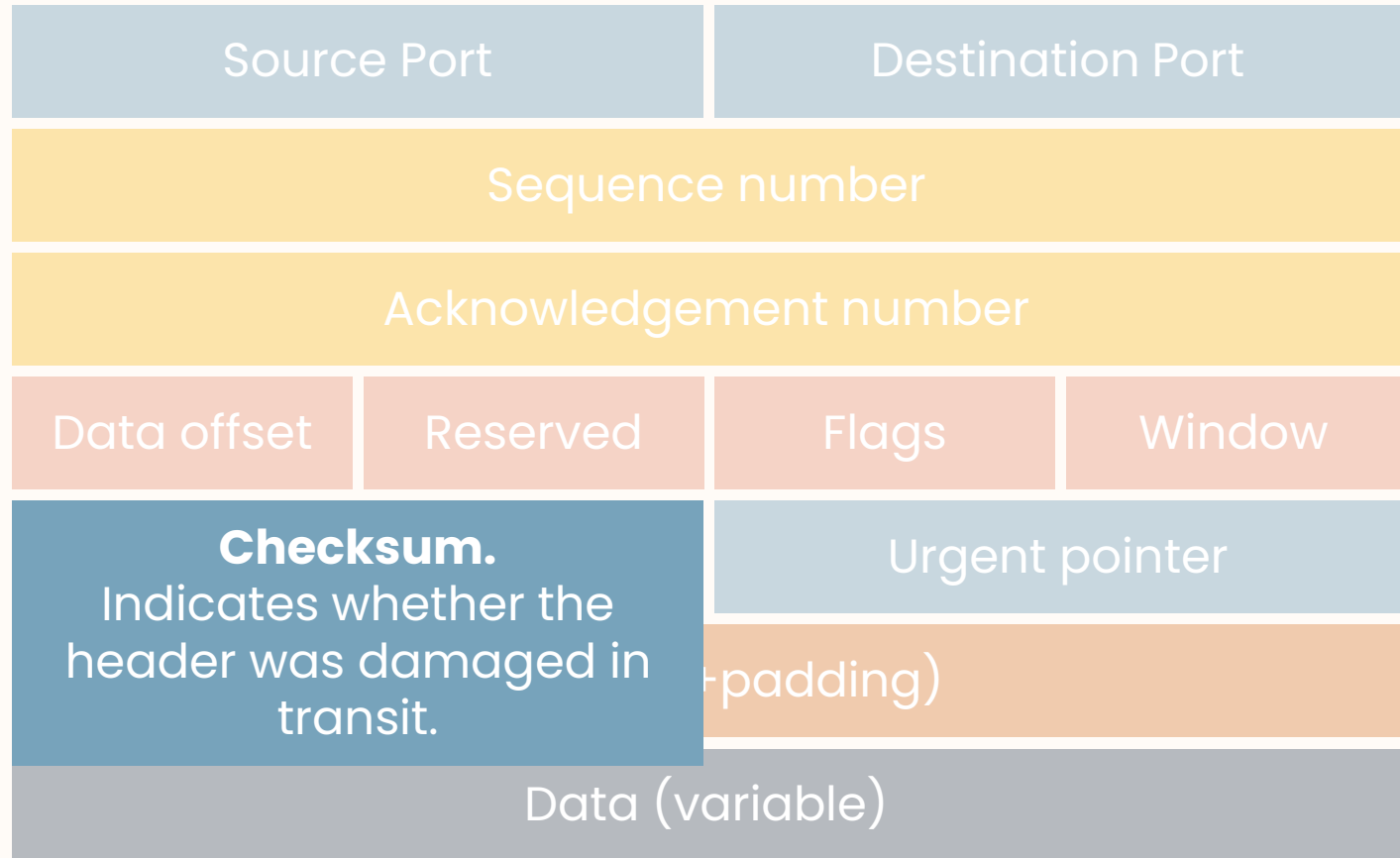
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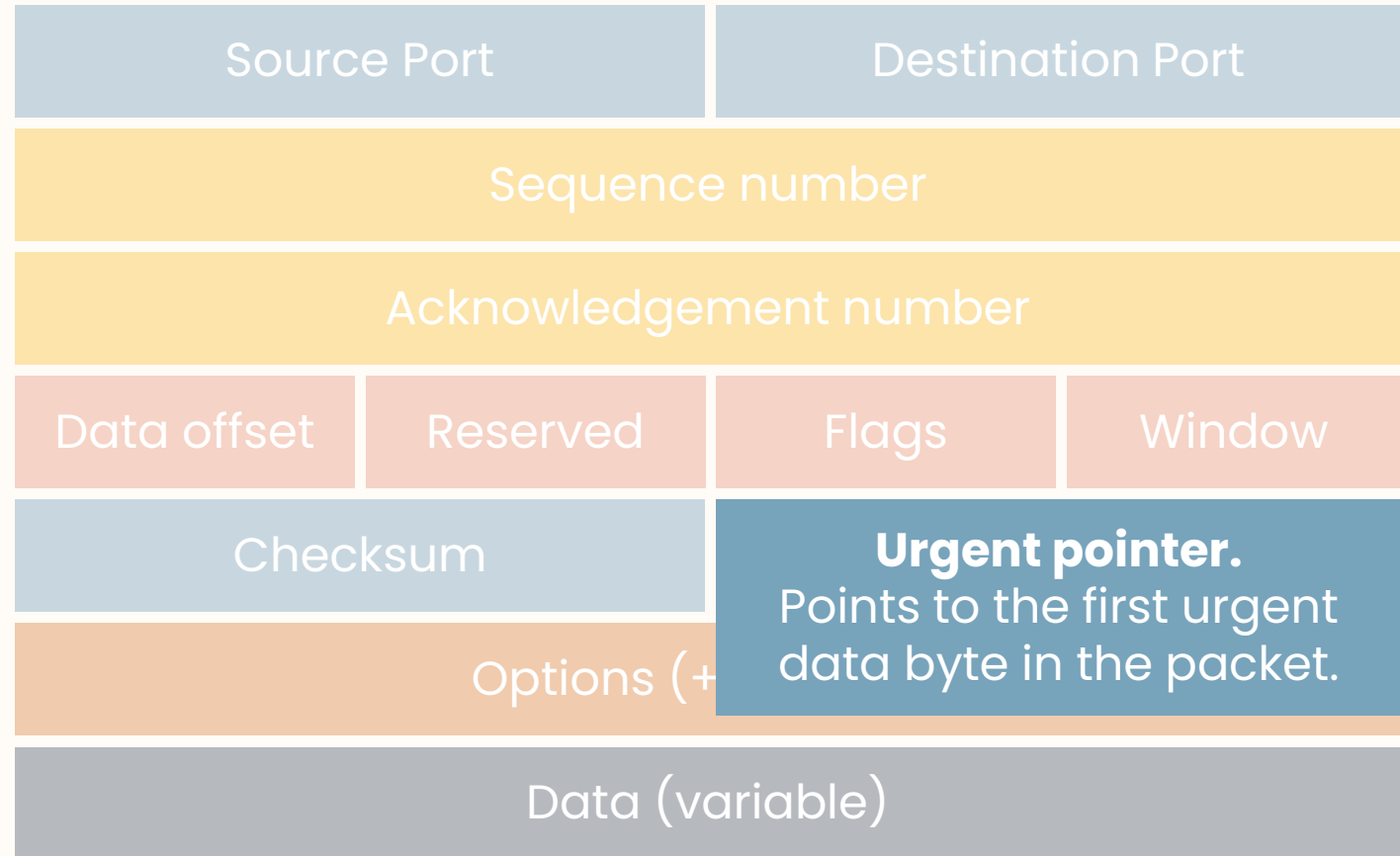
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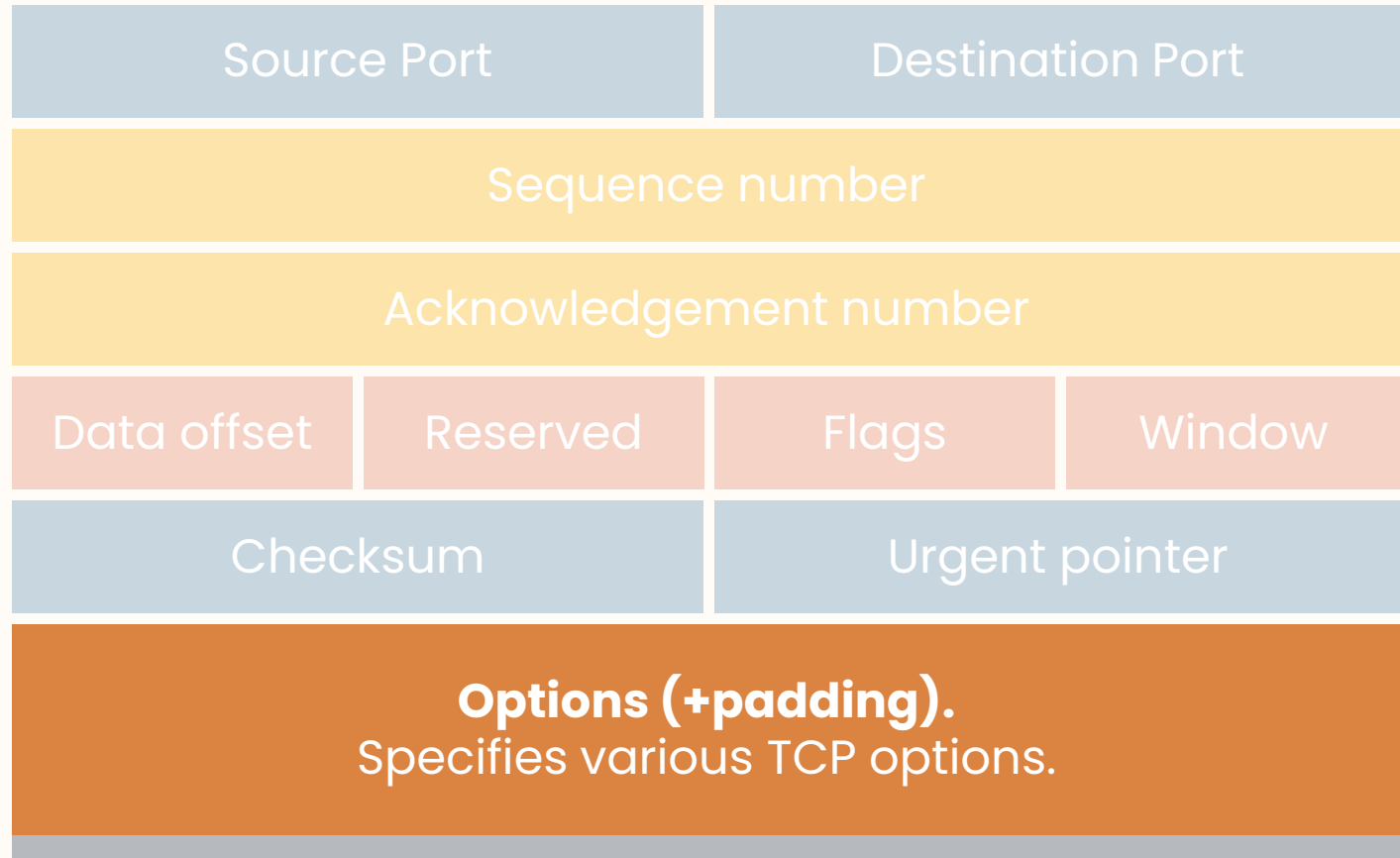
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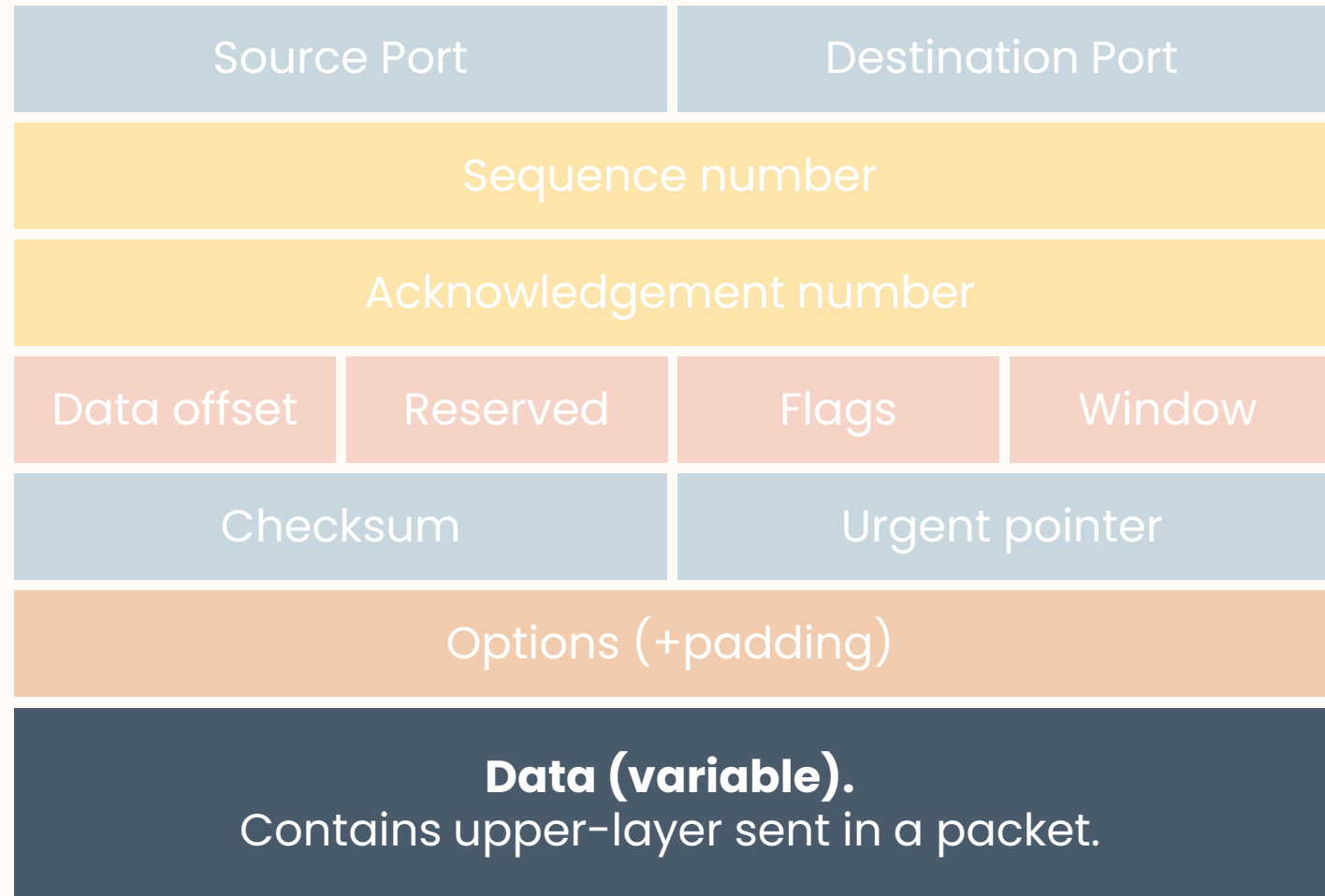
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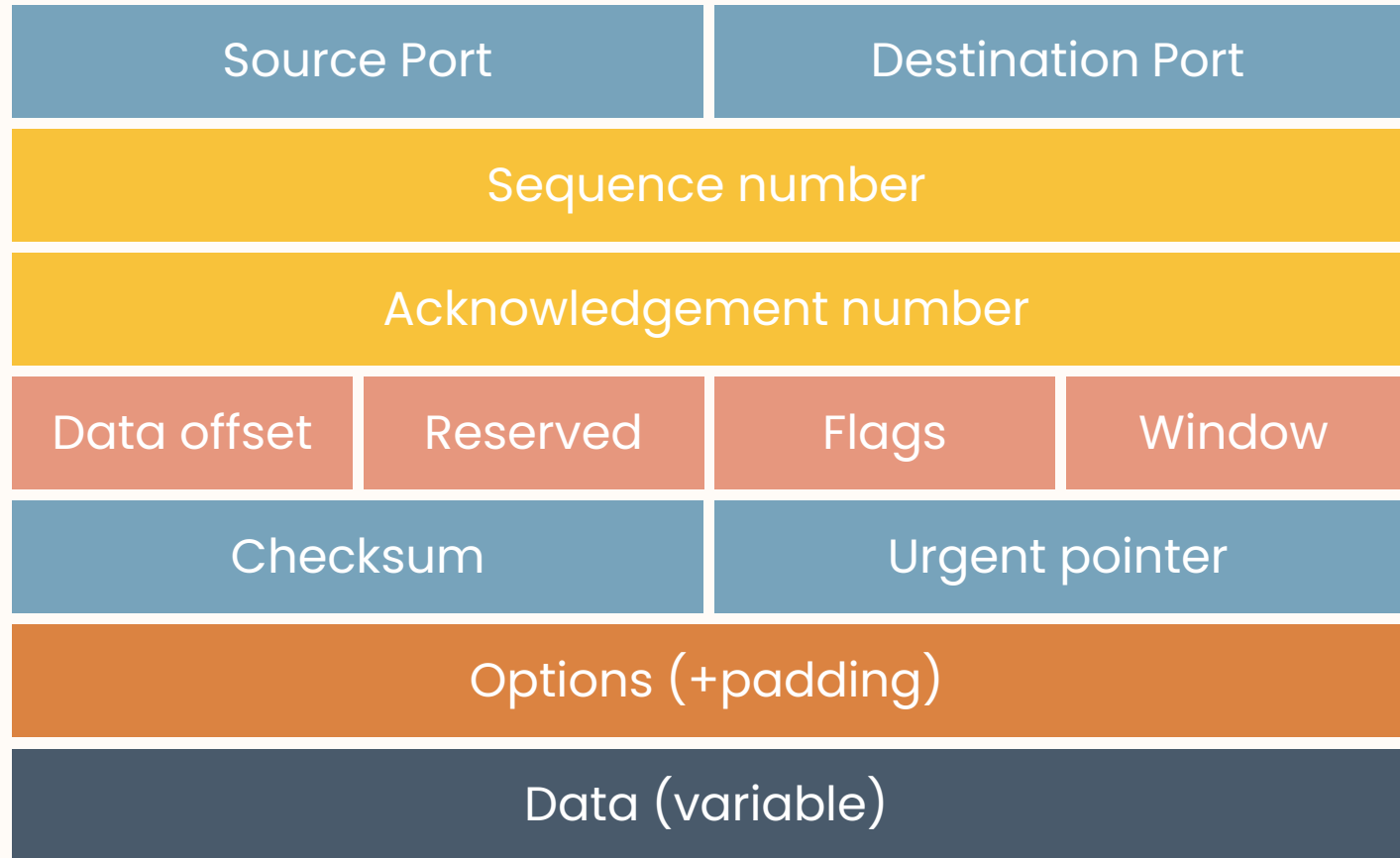
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Thank you!